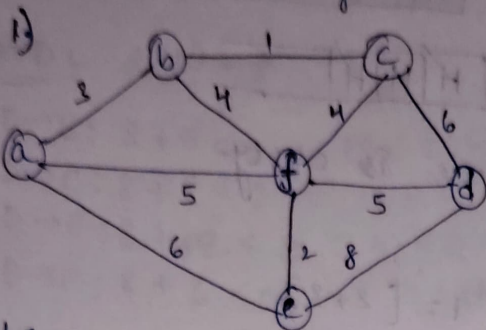


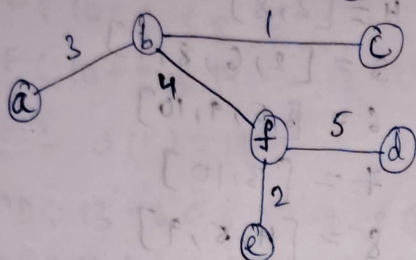
Kruskal's Algorithm (Minimum Spanning Trees)

A connected, acyclic subgraph of a connected graph G that contains all of G 's vertices, with n vertices it always has exactly $n-1$ edges.

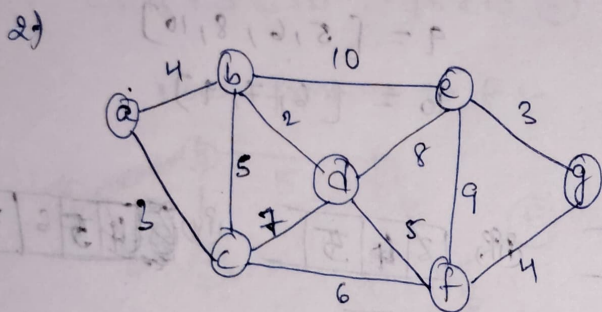


$$\frac{bc}{1} \quad \frac{ef}{2} \quad \frac{ab}{3} \quad \frac{bf}{4} \quad \frac{cf}{4} \quad \frac{af}{5} \quad \frac{df}{5} \quad \frac{ae}{6} \quad \frac{cd}{6} \quad \frac{de}{8}$$

(Note: In the original image, the edges cf , af , ae , cd , and de are marked with an 'X' to indicate they are rejected because they either create a cycle or result in a vertex with a degree greater than 2.)

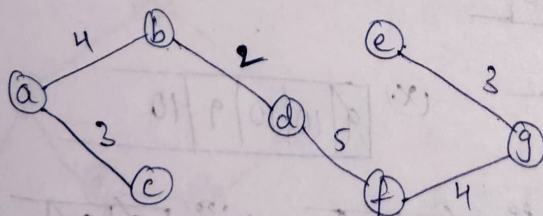


so, total weight of the MST = $1+2+3+4+5 = \underline{15}$



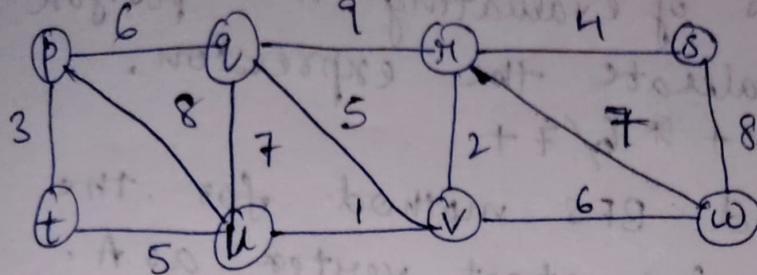
$$\frac{bd}{2} \quad \frac{ac}{3} \quad \frac{eg}{3} \quad \frac{ab}{4} \quad \frac{fg}{4} \quad \frac{bc}{5} \quad \frac{df}{5} \quad \frac{cf}{6} \quad \frac{cd}{7} \quad \frac{de}{8}$$

(Note: In the original image, the edges bc , cf , cd , and de are marked with an 'X' to indicate they are rejected because they either create a cycle or result in a vertex with a degree greater than 2.)

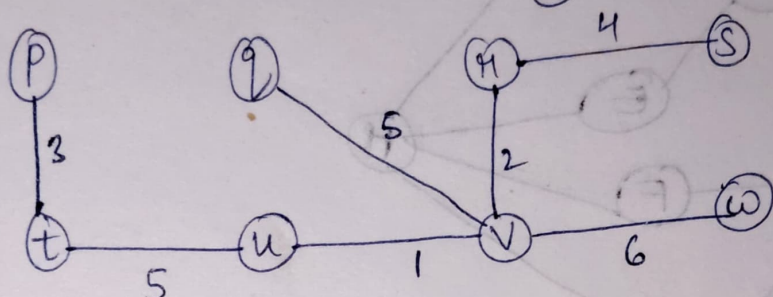


so, total weight of MST = $2+3+3+4+4 + 5 = \underline{21}$

3.)



- | | | | | | | | | | | |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| $\frac{uv}{1}$ | $\frac{uv}{2}$ | $\frac{pt}{3}$ | $\frac{rs}{4}$ | $\frac{qv}{5}$ | $\frac{tu}{5}$ | $\frac{pq}{6}$ | $\frac{vw}{6}$ | $\frac{qu}{7}$ | $\frac{rw}{7}$ | $\frac{pu}{8}$ |
| | | | | | | (X) | | (X) | (X) | (X) |



So, total weight of the MST = $1+2+3+4+5+8+6$
 $= 26$

→ If there is a cycle, do not consider that edge.

